An elementary determinant bound for the tenth discriminant

A proof with cutoff $q < 2^{15}$

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Abstract

We prove that an imaginary quadratic field of class number one has absolute fundamental discriminant $q < 32768$. A count of quadratic norms supplies an auxiliary inert prime, at which a derivative interpolation determinant is nonzero. Frobenius congruences at the inert primes then force more divisibility than its complex absolute value permits. The prime estimates are proved using factorials and binomial coefficients, and the numerical argument consists of one endpoint comparison. A separate integer sieve, using primes at most 31, excludes all remaining possibilities outside the nine classical discriminants. The proof is independent of the earlier versions in this project; its verification requires no numerical parameter search.

1 Introduction and arithmetic preliminaries

Tao's tenth-discriminant challenge [6] asks for explicit bounds extracted from recent arguments concerning Landau–Siegel zeros, including bounds that favor simple, human-verifiable proofs over numerical strength. We give such a bound here. The smaller cutoff $q < 570$ in [7] uses substantially more refinement; no result from that note is an input below.

For a negative fundamental discriminant D, put $q = |D|$ and $K = \mathbb{Q}(\sqrt{D})$.

Theorem 1. *If $h(K) = 1$, then $q < 32768 = 2^{15}$.*

The argument uses standard quadratic ideal theory, finite fields, Hadamard's inequality, and elementary analysis. All estimates specific to the proof are established below. The determinant is never computed: the supplementary code checks rational constants and a finite residue sieve.

We begin with the classical inertness obstruction emphasized by Silberman [5]. Every nonrational element of $\mathcal{O}_K$ has norm at least $q/4$, by its imaginary part in the usual integral basis. If $h(K) = 1$, a noninert rational prime p has a principal integral ideal of norm p. Its generator cannot be a rational integer, whose norm would be a square. Consequently

$$p < q/4 \quad \Longrightarrow \quad p \text{ is inert.} \tag{1}$$

For the proof of Theorem 1, suppose henceforth that $h(K) = 1$ and $q \geq q_0 := 32768$. Then q is odd, since otherwise 2 ramifies. It is also prime: an odd fundamental discriminant is squarefree, and a composite q would have a ramified prime factor at most $\sqrt{q} < q/4$. Inertness of 2 gives $q \equiv 3 \pmod 8$. Thus the only ramified prime is q, the units are ± 1. Put $u = \sqrt{q}$. Then

$$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\omega, \qquad \omega = (1 + iu)/2, \qquad Q(a, b) = |a + b\omega|^2 = (a + b/2)^2 + qb^2/4. \tag{2}$$

Write $L = \log q$; all logarithms are natural. We will use the elementary endpoint bounds

$$\frac{207}{20} < \log q_0 < \frac{21}{2}, \qquad \sqrt{q_0} > 180. \tag{3}$$

Indeed, $q_0 = 2^{15}$, $69/100 < \log 2 < 7/10$, and $180^2 < 32768$. The function $(\log q)/\sqrt{q}$ decreases for $\log q > 2$, so for every $q \geq q_0$,

$$L > 207/20, \qquad \frac{L}{u} < \frac{7}{120}. \tag{4}$$

2 Elementary prime estimates

Write

$$\vartheta(x) = \sum_{p \leq x} \log p, \qquad A(x) = \sum_{p \leq x} \frac{\log p}{p}, \qquad B = \sum_{p} \frac{\log p}{p^2}.$$

A letter p under a sum always denotes a rational prime. The following Chebyshev-style estimates suffice; no lower bound for ϑ is required.

Lemma 2. *For $x \geq 1$ and integers $m \geq 2$,*

$$\vartheta(x) < \frac{3}{2}x, \qquad \log m - \frac{7}{3} < A(m) < \log m + \frac{3}{2}, \qquad B < \frac{2}{3}. \tag{5}$$

Proof. Write $n = 2k$ or $n = 2k+1$. Every prime in $(\lceil n/2 \rceil, n]$ divides the central binomial coefficient $\binom{n}{\lfloor n/2 \rfloor}$. This coefficient is at most 2^{2k}: in the odd case there are two equal central coefficients. Thus

$$\vartheta(n) - \vartheta(\lceil n/2 \rceil) \leq 2\lfloor n/2 \rfloor \log 2.$$

Starting at $n = 1$, induction gives $\vartheta(n) < 2n \log 2$. Taking $n = \lfloor x \rfloor$, and using $\log 2 < 7/10 < 3/4$, proves the first assertion.

Every prime other than 2 is odd, and $f(x) = \log x / x^2$ decreases for $x \geq 3$. For each odd integer $n \geq 5$, $f(n) \leq \frac{1}{2} \int_{n-2}^{n} f(x)\, dx$. Hence

$$\begin{aligned} B &\leq \frac{\log 2}{4} + \frac{\log 3}{9} + \frac{1}{2} \int_3^\infty \frac{\log x}{x^2}\, dx \\ &= \frac{\log 2}{4} + \frac{5}{18} \log 3 + \frac{1}{6} < \frac{233}{360} < \frac{2}{3}, \end{aligned}$$

using $\log 3 < 11/10$.

The factorial valuation formula and integral comparison give

$$m \log m - m < \log(m!) \leq mA(m) + m \sum_p \frac{\log p}{p(p-1)} \leq mA(m) + 2mB < mA(m) + \frac{4}{3}m.$$

This proves the lower bound for $A(m)$. In the other direction, $\log(m!) \geq mA(m) - \vartheta(m)$ and $\log(m!) \leq m \log m$ prove the upper bound. □

3 Norm counts and an auxiliary inert prime

Pratt [4] suggested using class number one to bound split primes by norm representations, then using the resulting supply of inert primes in a determinant argument. We implement this with an unsieved norm count and a weighted interval of ordinary primes.

Let $\pi_+(x)$ count split rational primes at most x. For each integer $b > 0$, the real interval of a satisfying $Q(a,b) \le x$ has length $2\sqrt{x - qb^2/4}$, when nonempty. Bounding the number of integers by length plus 1, and the decreasing length sum by its integral, gives

$$\#\{(a,b) \in \mathbb{Z} \times \mathbb{Z}_{>0} : Q(a,b) \le x\} \le \frac{\pi x}{u} + \frac{2\sqrt{x}}{u}.$$

Each split prime has two generators with $b > 0$, one for each prime ideal above it: each ideal has generators $\pm\alpha$, exactly one of which has positive b. Consequently

$$\pi_+(x) \le \frac{\pi x}{2u} + \frac{\sqrt{x}}{u}. \tag{6}$$

For $X \ge Y \ge q/4$, partial summation, followed by (6), yields

$$\sum_{\substack{Y < p \le X \\ p \text{ split}}} \frac{\log p}{p} \le \frac{\pi}{4u}\left((\log X)^2 - (\log Y)^2 + 2\log Y\right) + \frac{2(\log Y + 1)}{u\sqrt{Y}}. \tag{7}$$

To see the boundary terms explicitly, start with

$$\frac{\log X}{X}\pi_+(X) - \frac{\log Y}{Y}\pi_+(Y) + \int_Y^X \pi_+(t)\frac{\log t - 1}{t^2}\,dt.$$

Discard the nonpositive lower boundary term. Substitution of (6) then gives the right side of (7), together with the further nonpositive term $-(\log X + 2)/(u\sqrt{X})$.

Put $S_+(X) = \sum_{p \le X,\ p \text{ split}} \log p/p$. There are no split primes at or below $q/4$, by (1) and the fact that $q/4$ is not an integer. Taking $Y = q/4$ in (7), and using $\log(q/4) > 2$ and $\pi < 16/5$, gives

$$S_+(X) \le \frac{4}{5u}(\log X)^2 + \frac{4(L+1)}{q} \qquad (X \ge q/4). \tag{8}$$

Lemma 3. *There is an inert prime R with $q^4 < R \le q^5$.*

Proof. Lemma 2 shows that the total weight $\log p/p$ of primes in $(q^4, q^5]$ is greater than $L - 23/6$. By (7), the split-prime weight there is at most

$$\frac{4}{5u}(9L^2 + 8L) + \frac{2(4L+1)}{uq^2}.$$

It is therefore enough to check

$$\frac{23}{6L} + \frac{4}{5u}(9L + 8) + \frac{2(4 + 1/L)}{uq^2} < 1. \tag{9}$$

Each term decreases as q increases in our range. For the middle term, differentiation with respect to L gives a factor $9 - (9L+8)/2 < 0$; the other terms plainly decrease. At $q = q_0$, (3) bounds the left side by

$$\frac{10}{27} + \frac{41}{90} + \frac{2(4 + 20/207)}{180 \cdot 32768^2} < \frac{10}{27} + \frac{41}{90} + \frac{1}{1000} < \frac{5}{6}.$$

The only ramified prime, q, is below the interval. Since the split primes cannot account for its full weight, an inert prime exists. □

4 The derivative determinant

We use Green's derivative matrix [1, §2], with a rectangular set of algebraic integers. Choose R as in Lemma 3 and put

$$S = \lfloor R/12 \rfloor, \qquad J = \lceil R/48 \rceil, \qquad N = RS, \qquad T = 2\sqrt{Ru}.$$

Take all points

$$z = a + b\omega, \qquad |a| \le A_0 := \lceil \sqrt{Ru} \rceil, \qquad |b| \le B_0 := \lceil \sqrt{R/u} \rceil.$$

Their number is $n = (2A_0 + 1)(2B_0 + 1) \ge 4R$, so $nJ \ge RS$. Since $|\omega| \le 3u/5$, $u > 1$, and $R > q^4 > 100u$,

$$|z| \le \frac{8}{5}\sqrt{Ru} + 1 + \frac{3}{5}u \le \frac{44}{25}\sqrt{Ru} < T.$$

Here $\sqrt{Ru} > 10u$. Also $2T < R$ and $1 < T < R$.

Form the matrix whose rows are indexed by (r, s), $0 \le r < R$, $0 \le s < S$, and whose columns are indexed by a point z_i and an order $0 \le k < J$. The entry is

$$\partial_z^k (z^r w^s)\Big|_{(z_i, \bar{z}_i)}. \tag{10}$$

These are ordinary derivatives, without division by factorials.

4.1 Nonvanishing over a finite field

The following argument is the auxiliary-prime method of van Dijk [8], highlighted in Green's comment [2]. Green also reports essentially the same method in unpublished work of Jacob Pescador; the present note uses the public argument and does not rely on that manuscript.

Reduce (10) modulo the inert prime ideal $R\mathcal{O}_K$. The residue field has R^2 elements. The nodes remain distinct: every nonzero element of $R\mathcal{O}_K$ has modulus at least R, because a nonzero algebraic integer has positive integral norm, whereas any difference of two nodes has modulus less than $2T < R$. Frobenius gives $\bar{z}_i = z_i^R$ in this field.

If a nonzero row combination vanished, let $F(z, w)$ be its polynomial and substitute $w = z^R$. The exponents $r + Rs$ are distinct, so $F(z, z^R)$ is nonzero and has degree at most $RS - 1$. Its derivatives agree with the prescribed partial derivatives, because the derivative of z^R is zero in characteristic R. Since $J \le R$ and $k! \ne 0$ for $0 \le k < J$, each node is a root of multiplicity at least J. This contradicts $nJ \ge RS$.

The matrix therefore has full row rank. Fix one nonzero N-square minor $D_0 \in \mathcal{O}_K$, and set

$$\Delta = D_0 \overline{D_0} \in \mathbb{Z}_{>0}.$$

Only this minor is used from now on.

4.2 The complex absolute-value bound

Each entry in row (r, s) has modulus at most $T^{r+s}(R/T)^J$: the derivative contributes at most R^k, and an entry with $k > r$ is zero. Hadamard's inequality gives

$$\frac{\log \Delta}{R^2 S} \le \frac{\log N}{R} + \left(1 + \frac{S}{R}\right) \log T + \frac{2J}{R} \log(R/T).$$

Now $S/R \le 1/12$, $J/R \le 1/48 + 1/R$, and $N < R^2$. Thus the coefficient of $\log T$, after combining the main terms, is $25/24$. Since $\log T = \log 2 + \frac{1}{2} \log R + \frac{1}{4} L$, we obtain

$$\frac{\log \Delta}{R^2 S} \le \frac{9}{16} \log R + \frac{25}{96} L + \frac{25}{24} \log 2 + \frac{4 \log R}{R}. \tag{11}$$

4.3 Divisibility at the inert primes

We use Green's triangular Frobenius transformation [1, Lemma 5.1], the stronger divisibility input suggested by Pratt [4]. The argument below applies also to $p = 2$. For an inert prime p, put

$$h_{r,s} = \min(\lfloor r/p \rfloor, S - 1 - s)$$

and replace each row polynomial $z^r w^s$ by

$$z^{r-ph_{r,s}}(z^p - w)^{h_{r,s}} w^s.$$

The leading term is $z^r w^s$, and every other term has smaller z-degree while remaining in the rectangle of row indices. Ordering rows by z-degree makes this an integer triangular transformation of determinant one.

At each node $z_i^p - \bar{z}_i \in p\mathcal{O}_K$, by Frobenius. Every positive ordinary derivative of $z^p - w$ is also divisible by p. By the product rule, all entries in the transformed row are therefore divisible by $p^{h_{r,s}}$. Taking norms gives

$$v_p(\Delta) \geq 2 \sum_{r,s} h_{r,s}. \tag{12}$$

The row transformation may depend on p; each divisibility is nevertheless a statement about the same integer Δ.

We bound this sum with generous constants. Direct summation gives, for every integer $k \geq 0$,

$$\sum_{s<S} \min(k, S - 1 - s) \geq Sk - \frac{k(k+1)}{2}.$$

Writing $R = ap + b$, $0 \leq b < p$, gives

$$\sum_{r<R} \lfloor r/p \rfloor = \frac{R(R-p) + b(p-b)}{2p} \geq \frac{R(R-p)}{2p}.$$

Also $\lfloor r/p \rfloor(\lfloor r/p \rfloor + 1) \leq r^2/p^2 + r/p$. Substituting these estimates yields

$$\frac{2\sum_{r,s} h_{r,s}}{R^2 S} \geq \left(1 - \frac{R-1}{2RS}\right)\frac{1}{p} - \frac{1}{R} - \frac{(R-1)(2R-1)}{6RSp^2}. \tag{13}$$

Let $I = \sum_{p \leq R,\ p \text{ inert}} \log p/p$. Summing (13), weighted by $\log p$, and using $S \geq R/13$ and Lemma 2, we find

$$\begin{aligned} \frac{\log \Delta}{R^2 S} &\geq I - \frac{I}{2S} - \frac{\vartheta(R)}{R} - \frac{R}{3S}B \\ &> I - \frac{I}{2S} - \frac{3}{2} - \frac{26}{9} = I - \frac{I}{2S} - \frac{79}{18}. \end{aligned} \tag{14}$$

Here $S \geq R/12 - 1 \geq R/13$, since $R > 156$.

5 One endpoint comparison

Separating the three splitting types in $A(R)$, and noting that the only ramified contribution is L/q, gives

$$I > \log R - \frac{7}{3} - S_+(R) - \frac{L}{q}, \qquad I < \log R + \frac{3}{2}.$$

Combine this with (11), (14), and (8). If

$$G(x) = \frac{7}{16}x - \frac{25}{96}L - \frac{4}{5u}x^2,$$

the resulting necessary inequality is

$$G(\log R) < \frac{121}{18} + \frac{25}{24}\log 2 + E, \qquad E = \frac{5L+4}{q} + \frac{I}{2S} + \frac{4\log R}{R}. \tag{15}$$

All boundary and rounding terms are included in E. In fact

$$E < \frac{5L+4}{q} + \frac{(21/2)\log R + 39/4}{R} < \frac{113}{65536} + \frac{1719}{4\cdot 10^{16}} < \frac{1}{500}. \tag{16}$$

For the second inequality, the first function decreases with q in our range, so use (3) at $q = q_0$. The second decreases with R; use $R > q_0^4 > 10^{16}$ and $\log(10^{16}) < 40$. Thus $\log 2 < 7/10$ gives

$$G(\log R) < \frac{121}{18} + \frac{35}{48} + \frac{1}{500} < \frac{15}{2}. \tag{17}$$

On the other hand, $4L < \log R \le 5L$ by Lemma 3, and G is concave. Its minimum on $[4L, 5L]$ is therefore at an endpoint. By (4),

$$G(5L) - G(4L) = L\left(\frac{7}{16} - \frac{36}{5}\frac{L}{u}\right) > \frac{7}{400}L > 0.$$

It follows that

$$G(\log R) \ge G(4L) = \frac{143}{96}L - \frac{64}{5}\frac{L^2}{u}.$$

Together with (17), this forces

$$\frac{143}{96} < \frac{64}{5}\frac{L}{u} + \frac{15}{2L} < \frac{56}{75} + \frac{50}{69} = \frac{846}{575} < \frac{143}{96},$$

a contradiction. This proves Theorem 1.

6 The remaining finite range

The residue sieve is the elementary small-prime method discussed by Pratt [3]. For $q > 8$, (1) at 2 forces $q \equiv 3 \pmod 8$, without requiring the large-discriminant assumption used above. For each odd prime $p < q/4$, inertness requires $-q$ to be a nonzero nonsquare modulo p.

There are 4095 integers $q \equiv 3 \pmod 8$ with $8 < q < 32768$. Test all of them, including nonfundamental values, against the primes

$$3, 5, 7, 11, 13, 17, 19, 23, 29, 31.$$

The following complete integer computation returns exactly the list shown in its assertion:

```
primes = (3, 5, 7, 11, 13, 17, 19, 23, 29, 31)
squares = {p: {a*a % p for a in range(p)} for p in primes}
survivors = [q for q in range(11, 32768, 8)
             if all(4*p >= q or (-q) % p not in squares[p]
                    for p in primes)]
assert survivors == [11, 19, 43, 67, 163]
```

The negative fundamental discriminants with $q \le 8$ have absolute values $3, 4, 7, 8$. We have therefore proved, with the finite computation just specified:

Corollary 4. *If $D < 0$ is a fundamental discriminant and $h(D) = 1$, then*

$$|D| \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

As an additional check of the familiar converse, the supplied script enumerates primitive reduced positive definite forms of each of these discriminants. Using $|b| \le a \le c$ and $b \ge 0$ when $|b| = a$ or $a = c$, it finds one form in each case. This uses the classical correspondence between reduced forms and ideal classes; it is separate from the exclusion argument.

Reproducibility

The supplementary files use only the Python standard library. Run

```
python3 verify_constants.py
python3 verify_finite.py
```

The first checks the numerical inequalities by rational arithmetic. For logarithms it uses 16 terms of

$$\log y = 2 \sum_{j \ge 0} \frac{t^{2j+1}}{2j+1}, \qquad t = \frac{y-1}{y+1}, \qquad 1 \le y \le 2,$$

with tail at most $2t^{33}/[33(1-t^2)]$, after reduction by powers of 2. The elementary identity $\pi/4 = \arctan(1/2) + \arctan(1/3)$, with three terms of each alternating series, verifies $\pi < 16/5$. These also verify $69/100 < \log 2 < 7/10$, $\log 3 < 11/10$, and $\log 10 < 5/2$, the only logarithmic inputs to the proof.

The second script performs the exhaustive residue check, verifies its residue tables by Euler's criterion, and enumerates the nine reduced-form lists. The scripts and their recorded outputs accompany this note. The monotonicity arguments in the text cover the infinite range; the scripts check its endpoints, not samples of that range. These computations are not formal verification, and independent mathematical review of this draft is pending.

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References

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